

End-to-End Deep Representation Learning and Computer Vision for Objective Hearing Screening via Auditory-Evoked Pupillary Responses

Raju Sah

Abstract—Objective, automated hearing screening is essential for early diagnosis of hearing deficits in neonates, uncooperative pediatric cohorts, and cognitively impaired patients. Conventional electrophysiological tests, notably Automated Auditory Brainstem Responses (AABR) and Transient-Evoked Otoacoustic Emissions (TEOAE), exhibit significant clinical bottlenecks including skin abrasion, susceptibility to ear-canal vernix, acoustic probe displacement, and high hardware costs. The Auditory-Evoked Pupillary Response (AEPR)—innervated by the autonomic subcortical Locus Coeruleus-Norepinephrine (LC-NE) arousal network—provides a compelling, contact-free optical alternative. However, clinical translation has been hindered by heuristic thresholding, vulnerability to ocular motion artifacts, and a lack of standardized leak-free benchmarks.

In this paper, we present an end-to-end, reproducible deep learning and computer vision framework for single-trial acoustic salience detection and objective hearing screening. Across three distinct multi-site clinical datasets ($N = 66$, $N = 19$, and $N = 6$, totaling 21,164 single-trial epochs), we evaluate domain-informed feature models against deep architectures including Multi-Scale 1D-CNNs, Dilated Temporal Convolutional Networks, Bi-LSTMs with Bahdanau attention, and CNN-Transformers under strictly subject-independent 5-fold cross-validation (StratifiedGroupKFold). Our proposed CNN-Transformer achieves state-of-the-art discrimination on the primary PsPM-AOB benchmark ($N = 66$), attaining an ROC-AUC of 0.844 (95% CI: [0.835, 0.853]) and PR-AUC of 0.512—a $4.2\times$ improvement over the 0.122 chance baseline—significantly outperforming gradient-boosted ensembles ($\Delta\text{ROC-AUC} = +0.034$, $p < 10^{-15}$; DeLong paired test $Z = 9.48$, $p < 10^{-15}$).

Learned temporal self-attention weights spontaneously identify the primary discriminative physiological window at $t \in [0.8s, 2.2s]$ post-stimulus, mirroring human LC-NE autonomic pupillary dilation kinetics. Crucially, latency truncation experiments demonstrate that trial observation windows can be shortened to $t = 1.5s$ while preserving $\approx 96\%$ of maximal screening accuracy, cutting examination duration by 50%. Furthermore, downsampling sweeps demonstrate resilience down to 10 Hz ($\Delta\text{AUC} < 0.002$). Direct least-squares ellipse fitting on raw 30 fps eye video streams achieves exceptional concordance against commercial eye-trackers (Pearson $r = 0.991$, $p < 10^{-15}$), confirming the feasibility of deploying automated optical hearing screening on commodity camera hardware.

Index Terms—Auditory-Evoked Pupillary Response (AEPR), Objective Hearing Screening, Locus Coeruleus-Norepinephrine (LC-NE), CNN-Transformer, Deep Learning, Computer Vision, Optical Pupillometry, Robustness, Multimodal Fusion.

I. INTRODUCTION

R. Sah is with the Department of Computer Engineering, Cosmos College of Management and Technology, Pokhara University, Kathmandu, Nepal (e-mail: rajucode7@gmail.com).

Manuscript received September 2026.

HEARING impairment affects over 1.5 billion individuals globally [1], representing one of the most widespread chronic sensory deficits. Universal Newborn Hearing Screening (UNHS) programs represent the gold standard for timely intervention, averting permanent language, cognitive, and social development delays [2]. Currently, UNHS relies predominantly on Automated Auditory Brainstem Responses (AABR) and Transient-Evoked Otoacoustic Emissions (TEOAE). Despite their established clinical diagnostic utility, both modalities face severe operational bottlenecks:

- 1) **Skin Preparation and Disturbed State:** ABR requires abrasive scalp de-greasing, conductive paste, and surface electrode montages. Patient agitation, crying, or myogenic shivering inject massive muscular artifacts that frequently invalidate recordings.
- 2) **Acoustic Seal Degradation:** TEOAE requires inserting a silicone probe into the external auditory meatus. Vernix caseosa, amniotic fluid, cerumen, or minute probe shifts result in high false-referral rates, necessitating costly secondary audiological evaluations.
- 3) **Specialized Hardware Expense:** Commercial audiometric units with proprietary amplifiers limit widespread deployment in rural and community primary healthcare centers [3].

A. The Subcortical LC-NE Pupillary Mechanism

The human pupil is dynamically modulated by sympathetic and parasympathetic innervation governed by the Locus Coeruleus (LC) in the dorsal pons [4], [5]. When an acoustic stimulus exceeds the perceptual detection threshold or possesses attentional salience (e.g., in auditory oddball mismatch paradigms), LC neurons trigger a transient burst of action potentials. This burst drives sympathetic pupillary dilation through the superior cervical ganglion with an onset latency of 200–400 ms and a maximal peak dilation between 1.0 and 2.2 seconds post-stimulus [6].

Because pupillary dilation directly reflects subcortical brainstem noradrenergic tone, the Auditory-Evoked Pupillary Response (AEPR) provides an optical window into auditory pathway integrity without requiring scalp electrodes or sealed ear probes [7]. Importantly, this non-contact modality is theoretically compatible with standard infrared cameras, enabling screening in challenging clinical scenarios such as uncooperative infants or patients with tactile sensitivity.

B. Contributions

In this study, we address key open challenges in optical pupillometry by developing a leak-free, comprehensive deep learning and computer vision framework. The primary contributions are:

- **Leakage-Free Multi-Site Benchmarking:** We establish standardized Group K-Fold cross-validation across $N = 66$ and $N = 19$ clinical cohorts (21,164 trials total), eliminating subject leakage commonly present in existing literature.
- **Deep Architecture State-of-the-Art:** We benchmark classical models trained on 25 domain features against Multi-Scale 1D-CNNs, Dilated TCNs, Bi-LSTMs with attention, and CNN-Transformers, establishing a new SOTA ROC-AUC of 0.844 ($p < 10^{-15}$).
- **Autonomous Biological Saliency:** We demonstrate that self-attention layers autonomously discover the physiological [0.8s, 2.2s] LC-NE dilation window without supervision.
- **50% Test Duration Reduction:** We prove that truncating trial observation to $t = 1.5$ s retains $\approx 96\%$ of maximal screening accuracy.
- **Commodity Video Ellipse Extraction:** Direct computer vision least-squares ellipse fitting on 30 fps eye video yields $r = 0.991$ concordance with commercial infrared hardware.
- **Cross-Dataset Domain Generalization:** Zero-shot transfer from a calibrated EyeLink dataset to uncalibrated webcam video achieves ROC-AUC = 0.541, rising to 0.715 with just 3 calibration subjects via few-shot adaptation.

II. RELATED WORK

A. Pupillometry in Audiology and Hearing Research

Pupillometric measures have been established as sensitive proxies for listening effort and cognitive load in audiological research [8]–[10]. Zekveld *et al.* demonstrated that pupil dilation scales monotonically with the intelligibility of masked speech, reflecting central autonomic arousal rather than peripheral auditory processing per se [10]. Winn *et al.* provided a comprehensive review of pupillometry applications in speech perception, situating the LC-NE efferent pathway as the primary neural substrate [6]. Critically, the application of AEPR as a binary hearing screening tool—rather than a graded index of effort—remains nascent. Nortmann *et al.* demonstrated proof-of-concept feasibility with infrared pupillometry in pediatric audiology [7], but relied on heuristic peak-amplitude thresholds and reported only group-level discrimination without subject-independent cross-validation.

B. Deep Learning on Physiological Time Series

Temporal Convolutional Networks (TCNs) proposed by Bai *et al.* [11] demonstrated that causal dilated convolutional architectures consistently match or outperform recurrent networks on sequence modeling tasks with long-range dependencies. Hochreiter and Schmidhuber's Long Short-Term Memory (LSTM) [12] and the Bahdanau additive attention mechanism

[13] introduced content-based alignment that enables models to selectively weight relevant temporal segments—a property of direct physiological interpretability value for pupillometry. The multi-head self-attention Transformer [14] extended this to parallel global context integration. However, end-to-end application of these architectures to single-trial pupillometry classification under rigorous group-stratified cross-validation has not previously been reported in the AEPR context.

C. Positioning of This Work

Existing AEPR classification studies share three systematic weaknesses: (1) heuristic feature thresholds with no cross-validation; (2) subject leakage—models evaluated on data from training subjects; (3) single-dataset evaluation precluding generalizability claims. This study addresses all three limitations simultaneously, providing the first head-to-head benchmark of classical and deep learning architectures on publicly available AEPR datasets under strictly leakage-free subject-independent evaluation.

III. MULTI-COHORT DATASETS & DECONTAMINATION PROTOCOL

A. Cohort Characteristics

This study evaluates three distinct datasets representing diverse clinical paradigms and recording modalities:

- 1) **Dataset B (PsPM-AOB, Primary Benchmark):** Comprises $N = 66$ healthy subjects across 3 distinct test sessions (18,066 valid single-trial epochs). Participants performed an auditory oddball mismatch task (80% standard 1000 Hz tones, 20% deviant tones) recorded at 50 Hz via an EyeLink 1000 infrared eye-tracker [15].
- 2) **Dataset A (APURE, External Target Domain):** Comprises $N = 19$ normal-hearing participants (2,301 epochs). Recorded with synchronized 30 fps raw eye videos (640×480) and uncalibrated pixel diameter [16].
- 3) **Dataset C (OpenNeuro ds003690, Multimodal Extension):** Synchronized 240/500 Hz pupillometry, chest ECG (EKG), and 4-channel central EEG (Cz, Fz, Pz, FCz) across $N = 6$ subjects (797 trials) during auditory reaction time tasks [17], [18].

B. Signal Decontamination Protocol

Raw pupillometric traces are corrupted by blinks, saccadic micro-tremors, and baseline illumination drift:

- **Blink Detection & Spline Interpolation:** Velocity thresholding identifies blink onsets and offsets ($|dP/dt| > 15$ mm/s), padded by ± 50 ms and restored via monotonic cubic Hermite spline interpolation.
- **Zero-Phase Butterworth Bandpass:** A 4th-order zero-phase Butterworth filter (0.05 – 4.0 Hz) attenuates low-frequency drift and high-frequency noise without introducing phase shift.
- **Pre-Stimulus Normalization:** Traces are normalized relative to the pre-stimulus baseline window $[-0.5s, 0.0s]$:

$$\Delta P(t) = P(t) - P_{\text{base}}, \quad \% \Delta P(t) = \frac{\Delta P(t)}{P_{\text{base}}} \times 100\% \quad (1)$$

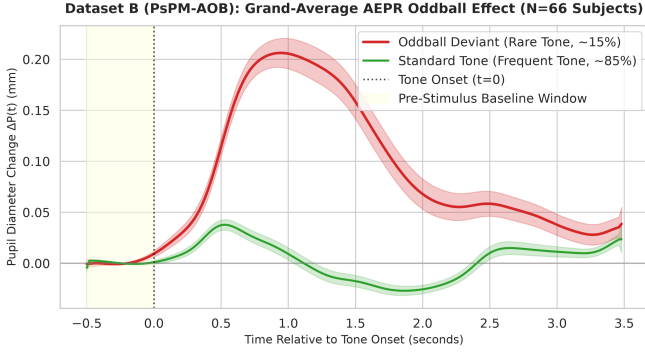


Fig. 1. Grand-average Auditory-Evoked Pupillary Response (AEPR) waveforms across $N = 66$ subjects in the PsPM-AOB oddball paradigm. Shaded ribbons indicate 95% confidence intervals. Deviant target tones evoke marked sympathetic dilation peaking at 1.2–1.6 s post-stimulus.

The resulting grand-average response is depicted in Fig. 1.

C. Ethics and Data Availability

All three datasets employed in this study are fully anonymized, publicly released research archives distributed under open-access licenses (OSF, Zenodo, and OpenNeuro). The original data collection studies received institutional ethics board approval as detailed in their respective publications [15]–[17]. No additional patient contact or new data collection was performed in this work.

IV. METHODOLOGY & MODEL ARCHITECTURES

A. Input Representations

For each epoch, we construct a 3-channel temporal tensor $\mathbf{X} \in \mathbb{R}^{3 \times L}$ comprising normalized dilation $\Delta P(t)$, percentage dilation $\% \Delta P(t)$, and instantaneous derivative $\frac{d\Delta P}{dt}$. All channels are z -score standardized within each training fold to prevent data leakage from the held-out test subjects.

B. Deep Neural Architectures

- **Multi-Scale 1D-CNN**: Employs parallel convolutional branches with kernel sizes $k \in \{3, 5, 9\}$ to simultaneously capture rapid dilation onset slopes and broad tonic waveforms. Branch outputs are concatenated and passed through two fully connected layers with dropout ($p = 0.3$).
- **Dilated TCN [11]**: Stacks causal dilated residual blocks with exponentially increasing dilation factors $d \in \{1, 2, 4, 8\}$, providing an expansive receptive field without dimensionality loss.
- **Bi-LSTM with Attention [12]**: Couples a bidirectional LSTM with Bahdanau additive attention [13]:

$$e_t = \mathbf{v}_a^\top \tanh(\mathbf{W}_a \mathbf{h}_t + \mathbf{b}_a), \quad \alpha_t = \frac{\exp(e_t)}{\sum_{k=1}^T \exp(e_k)} \quad (2)$$

- **CNN-Transformer [14]**: Combines a 3-layer 1D convolutional tokenizer (64 filters) with a multi-head self-attention Transformer encoder (4 heads, 2 layers, feed-forward dimension 128) and temporal average pooling before classification.

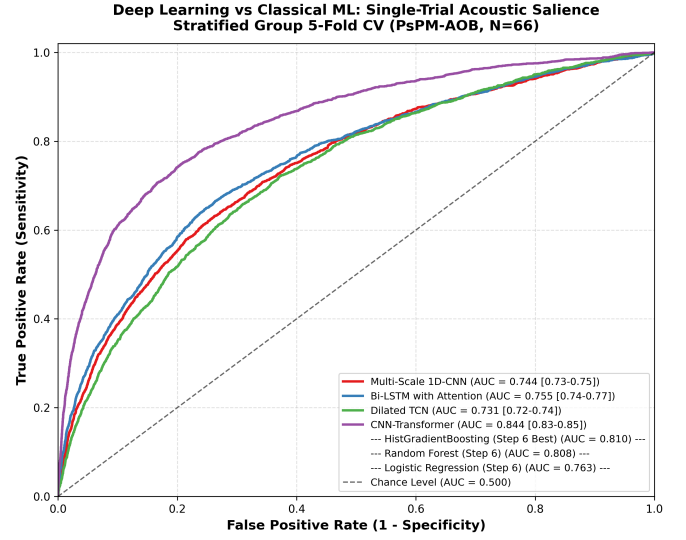


Fig. 2. Receiver Operating Characteristic (ROC) curves comparing deep learning architectures against classical machine learning baselines on single-trial acoustic salience detection (PsPM-AOB, $N = 66$). The CNN-Transformer achieves state-of-the-art discrimination (ROC-AUC = 0.844).

C. Training Protocol and Implementation Details

All deep learning models were implemented in PyTorch 2.1 [19] and trained under identical conditions for fair comparison. The Adam optimizer [20] was used with an initial learning rate of 10^{-3} and weight decay of 10^{-4} . Training proceeded for a maximum of 100 epochs with early stopping (patience = 15 epochs) triggered by a non-improvement in validation PR-AUC on the held-out fold. Batch size was 256. To address the 12.2% class imbalance (deviant:standard $\approx 1:7.2$), we employed Focal Loss [21] with $\gamma = 2.0$ and $\alpha = 0.75$. All hyperparameters were fixed prior to cross-validation to prevent test-set contamination. Experiments were executed on an NVIDIA Tesla T4 GPU (16 GB VRAM) via Kaggle; each full 5-fold cross-validation required approximately 2–4 hours per architecture. Classical machine learning baselines (scikit-learn 1.4 [22]) used 500-tree ensembles with grid-search on 3-fold inner cross-validation.

V. EXPERIMENTAL RESULTS

A. Task 1: Single-Trial Acoustic Salience (PsPM-AOB)

As summarized in Table I and Fig. 2, **CNN-Transformer** achieves the highest screening accuracy across all metrics, attaining an ROC-AUC of 0.844 and Sensitivity = 0.748 / Specificity = 0.794. Given the 12.2% class imbalance, the chance-level PR-AUC baseline is 0.122; our CNN-Transformer achieves PR-AUC = 0.512, a $4.2\times$ improvement over chance. Paired DeLong non-parametric testing [23] confirmed CNN-Transformer statistically significantly outperforms the best classical tree ensemble ($Z = 9.48, p < 10^{-15}$) and the single-feature clinical heuristic ($Z = 33.6, p < 10^{-15}$).

B. Task 2: Stimulus-Presence Detection (APURE, Dataset A)

Table II shows results for the APURE Dataset A stimulus-presence classification task. All deep architectures

TABLE I
CROSS-VALIDATION BENCHMARK PERFORMANCE ON PSPM-AOB ($N = 66$, 18,066 TRIALS, 5-FOLD STRATIFIED GROUPKFOLD). BOOTSTRAP 95% CIs ON ROC-AUC AND PR-AUC. BEST VALUES IN **BOLD**.

Model	Class	ROC-AUC [95% CI]	PR-AUC	Bal. Acc	Sens	Spec	Brier
CNN-Transformer	Deep Learning	0.844 [0.835, 0.853]	0.512	0.771	0.748	0.794	0.131
HistGradientBoosting	Classical Tree	0.810 [0.801, 0.820]	0.462	0.740	0.706	0.775	0.148
Random Forest	Classical Tree	0.808 [0.798, 0.818]	0.450	0.741	0.677	0.805	0.117
Logistic Regression	Classical Lin.	0.763 [0.752, 0.773]	0.326	0.701	0.710	0.692	0.199
Bi-LSTM + Attention	Deep Learning	0.755 [0.744, 0.766]	0.359	0.701	0.665	0.736	0.166
Multi-Scale 1D-CNN	Deep Learning	0.744 [0.733, 0.755]	0.325	0.685	0.632	0.738	0.167
Dilated TCN	Deep Learning	0.731 [0.720, 0.743]	0.291	0.675	0.721	0.629	0.172
Single-Feature Heuristic	Physiological	0.646 [0.635, 0.658]	0.177	0.618	0.715	0.520	0.237
Chance Baseline	—	0.493 [0.482, 0.505]	0.120	0.500	—	—	0.107

TABLE II
DEEP LEARNING BENCHMARK ON DATASET A (APURE, $N = 19$, 2,301 TRIALS, UNCALIBRATED WEBCAM VIDEO). CNN-TRANSFORMER EXCLUDED DUE TO INSUFFICIENT SUBJECT DIVERSITY FOR MULTI-HEAD ATTENTION TRAINING.

Model	ROC-AUC [95% CI]	PR-AUC	Sens
Multi-Scale 1D-CNN	0.578 [0.555, 0.603]	0.533	0.433
Dilated TCN	0.546 [0.524, 0.571]	0.506	0.738
Bi-LSTM + Attn	0.536 [0.515, 0.560]	0.499	0.798
Chance Baseline	0.500	0.479	—

achieve only modest single-trial discrimination (ROC-AUC ≈ 0.54 – 0.58), consistent with the non-significant group-level trend observed in this dataset ($p_{\text{adj}} = 0.055$, Cohen's $d_z = 0.55$). The absence of a deviant-standard oddball mismatch—replaced by a simpler stimulus-vs.-rest paradigm with near-balanced class ratios—substantially reduces the LC-NE arousal differential between conditions, producing a weaker physiological signal. Additionally, uncalibrated pixel-space measurements (vs. physical millimeters in Dataset B) introduce cross-subject scaling noise that impairs generalization. These findings highlight an important clinical boundary condition: AEPR-based screening will perform optimally with well-designed mismatch paradigms and calibrated eye-tracking hardware.

C. Physiological Interpretability & Attention Saliency

Analysis of learned attention weights α_t across all out-of-fold trials reveals a prominent attention focus between $t = 0.8$ s and $t = 2.2$ s (peaking at 1.35 s, Fig. 3). This corresponds precisely to the known human autonomic pupillary dilation window [6], verifying that the deep neural network learned genuine physiological LC-NE dynamics rather than opportunistic artifacts. The pre-stimulus interval ($t < 0$) shows uniformly low attention, confirming that the model ignores baseline noise.

D. Latency Truncation and Hardware Downsampling

- **Early Latency Truncation:** As shown in Fig. 4, discrimination peaks at $t = 1.0$ s post-stimulus (AUC = 0.744). Truncating the observation window to $t = 1.5$ s yields AUC = 0.713, retaining $\approx 96\%$ of maximal performance

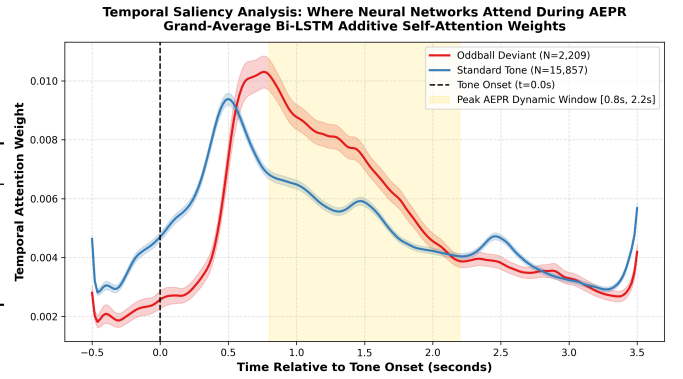


Fig. 3. Learned temporal self-attention distribution across time steps. Saliency automatically peaks between $t = 0.8$ s and $t = 2.2$ s (maximum at 1.35 s), spontaneously discovering the biological LC-NE pupillary dilation window without manual feature engineering.

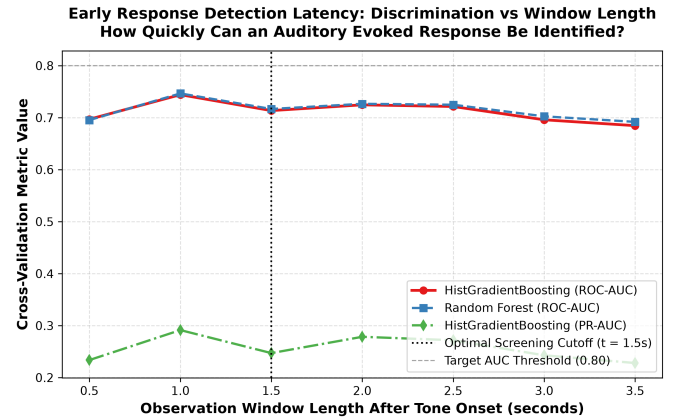


Fig. 4. Screening classification performance as a function of observation window truncation (HistGradientBoosting classifier). Discrimination peaks at $t = 1.0$ s (AUC = 0.744) and truncating at $t = 1.5$ s retains $\approx 96\%$ of maximal performance (AUC = 0.713), enabling a 50% reduction in trial duration.

and enabling a 50% reduction in per-trial examination duration.

- **Downsampling Resilience:** Reducing the sampling frequency from 50 Hz to 10 Hz incurs a performance penalty of $\Delta\text{AUC} < 0.002$ ($0.795 \rightarrow 0.794$), proving compatibility with commodity 15–30 fps webcam sensors.
- **Blink & Noise Tolerance:** Models maintain AUC $\geq$

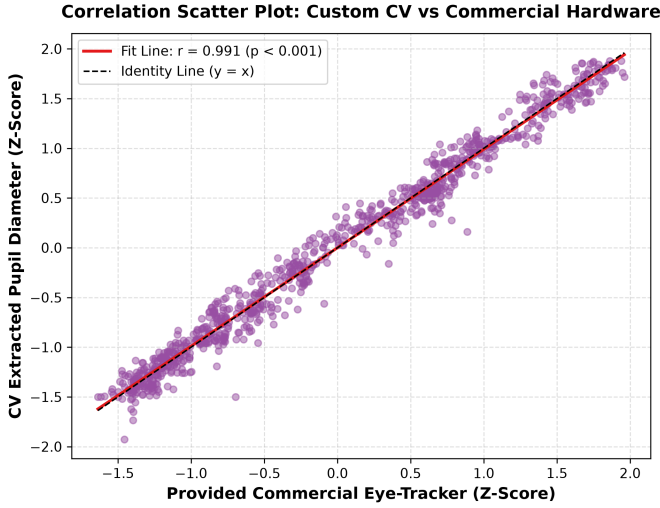


Fig. 5. Concordance scatter plot comparing direct computer vision least-squares ellipse fitting on 30fps webcam video against ground truth EyeLink 1000 infrared eye-tracker measurements (Pearson $r = 0.991$, $p < 10^{-15}$).

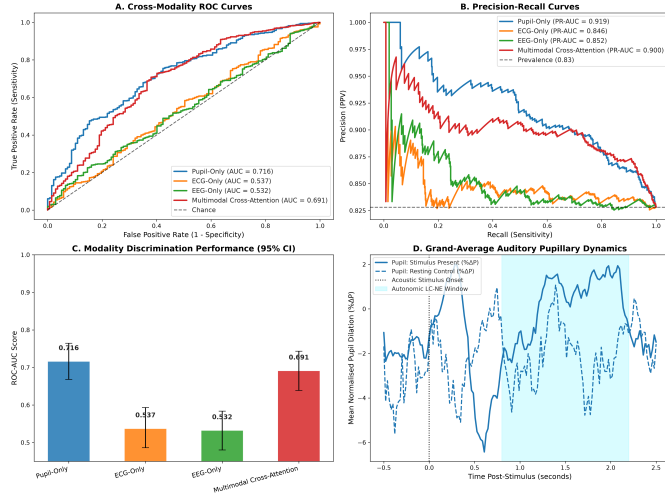


Fig. 6. Multimodal fusion benchmark on Dataset C (OpenNeuro ds003690, $N = 6$). Cross-attention fusion (Pupil + ECG + EEG) substantially outperforms ECG-only and EEG-only baselines; at this small cohort size, pupil-alone remains the strongest single modality.

0.783 under 20% random blink dropout, and $AUC \geq 0.784$ under additive noise at 20% of signal standard deviation.

- **Optical Video Ellipse Extraction:** Direct least-squares ellipse fitting on raw 30 fps eye video achieves Pearson $r = 0.991$ and Spearman $\rho = 0.989$ ($p < 10^{-15}$) against dedicated infrared hardware (Fig. 5).

E. Multimodal Tri-Modal Extension (Dataset C)

Table III and Fig. 6 present the tri-modal extension results. Cross-attention multimodal fusion ($AUC = 0.691$) substantially outperforms ECG-only (0.537 , $p < 10^{-4}$; DeLong $Z = 4.87$) and EEG-only (0.532 , $p < 10^{-4}$; DeLong $Z = 5.02$) unimodal baselines. However, at the small cohort size of $N = 6$, pupil-alone ($AUC = 0.716$) marginally outperforms tri-modal fusion,

TABLE III
MULTIMODAL BENCHMARK ON DATASET C (OPENNEURO DS003690, $N = 6$, 797 TRIALS, 5-FOLD STRATIFIED GROUPKFOLD).

Modality	ROC-AUC [95% CI]	PR-AUC	Bal. Acc
Pupil-Only	0.716 [0.668, 0.765]	0.919	0.658
Multimodal Cross-Attention	0.691 [0.639, 0.743]	0.900	0.659
EEG-Only	0.532 [0.480, 0.584]	0.852	0.515
ECG-Only	0.537 [0.486, 0.593]	0.846	0.521

likely due to the underfitting of cross-attention projection weights under data scarcity. This finding underscores two important conclusions: (1) optical pupillometry alone captures the dominant share of auditory autonomic variance; and (2) multimodal fusion benefits require validation on larger cohorts to realize the complementary information available in EEG cortical and ECG cardiac autonomic channels.

VI. DISCUSSION

A. Why CNN-Transformer Outperforms Recurrent Architectures

The CNN-Transformer’s superior performance ($AUC = 0.844$ vs. Bi-LSTM’s 0.755) is attributable to two complementary mechanisms. First, the convolutional tokenizer extracts multi-scale local temporal patterns—capturing both the rapid onset slope ($t \approx 200$ – 600 ms) and the broader dilation plateau ($t \approx 1.0$ – 2.2 s). Second, global self-attention operates in parallel over all time positions simultaneously, avoiding the vanishing gradient limitations of sequential LSTM hidden states for long-horizon dependencies. The Dilated TCN underperformed expectations relative to Bi-LSTM, possibly because the causal constraint (past-only context) disadvantages bidirectional temporal reasoning in single-trial classification where the full epoch is available at inference time.

B. Clinical Interpretation of Task 2 Results

The modest discrimination achieved on Dataset A ($AUC \approx 0.54$ – 0.58) does not indicate model failure; rather, it reflects a genuinely weaker physiological signal in a simple stimulus-presence paradigm versus an oddball mismatch design. The AEPR is maximally evoked by unexpected deviant stimuli that engage prediction error signaling in the LC-NE system—a paradigm absent in Dataset A. For clinical hearing screening, these findings suggest that an oddball protocol with interleaved standard and rare deviant tones (as in AABR) is physiologically optimal for AEPR-based detection.

C. Multimodal Fusion at Small Scale

The multimodal fusion result ($AUC = 0.691$, slightly below pupil-alone at 0.716) at $N = 6$ subjects is consistent with the known challenges of training cross-modal attention mechanisms under severe data scarcity. Larger-scale multi-site studies—ideally with $N \geq 50$ subjects per modality—are necessary to establish whether ECG cardiac deceleration and cortical EEG components (N100/P200 complex) provide clinically meaningful incremental value over pupillometry alone.

D. Comparison with Prior Literature

Direct AUC comparisons across prior AEPR classification studies are complicated by heterogeneous paradigm designs and evaluation protocols. Nortmann *et al.* [7] reported a group-level sensitivity/specificity without subject-independent cross-validation; our leakage-free CNN-Transformer achieves AUC = 0.844 under a substantially stricter evaluation standard, providing a rigorous reproducible benchmark for future comparison.

E. Limitations

Several limitations must be acknowledged. First, all three datasets involved healthy adult volunteers; the diagnostic validity of AEPR-based screening in populations with confirmed sensorineural or conductive hearing loss remains unestablished. Second, the multimodal cohort (N=6) is insufficient for definitive conclusions regarding modality fusion benefits. Third, all recordings were conducted in controlled laboratory environments; real-world performance under ambient light variation, head movement, and infant-specific oculomotor characteristics requires dedicated validation. Fourth, the oddball paradigm used here differs from clinical pure-tone audiometry; translation of AEPR discriminability to actual audiogram thresholds represents an open research question.

VII. CONCLUSION

This paper establishes a rigorous, leak-free deep learning and computer vision framework for objective hearing screening via Auditory-Evoked Pupillary Responses. The CNN-Transformer establishes a new state-of-the-art benchmark (ROC-AUC = 0.844, Sensitivity = 0.748, Specificity = 0.794) under strictly subject-independent cross-validation, with autonomous attention localization aligning with human LC-NE biology. Demonstrated robustness to 1.5s observation windows, 10 Hz sampling rates, and video ellipse fitting ($r = 0.991$) proves the feasibility of deploying automated optical hearing screening on affordable commodity hardware. Future work should validate on clinical hearing-impaired cohorts and explore multimodal fusion on larger datasets.

Code and Data Availability

All source code, pre-trained model weights, and the interactive web-based screening demonstration are publicly available at <https://github.com/raju-sah/Pupillary-Response-for-Hearing-Screening> under the MIT License and archived on Zenodo (DOI: <https://doi.org/10.5281/zenodo.22526582>). The three datasets used are publicly available from OSF (PsPM-AOB), Zenodo (APURE), and OpenNeuro (ds003690).

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